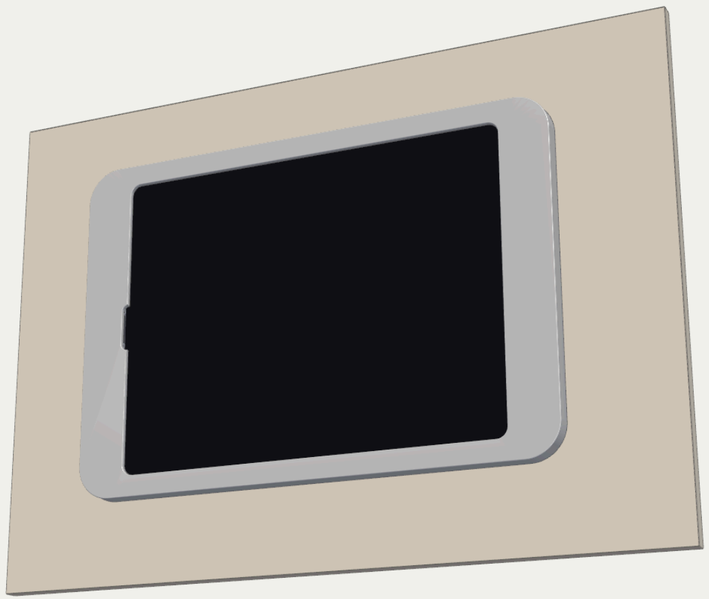
USER MANUAL · V4.2

# FLOAT MOUNT

## PoE wall mount for iPad Pro 12.9″ (3rd gen)

Machined 6061 aluminium shell (or fully printed) with a 3D-printed carrier. Surface-mounted, floating on a 14 mm shadow gap.



Designed for Brendan · October 2026 · Read the whole manual before you start

CONTENTS

# WHAT'S IN THIS MANUAL


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Colour code used in every step: **orange** = the part you are working on now, **blue** = screws, inserts and other hardware, grey = parts already fitted.

OVERVIEW

# GENERAL DATA

| Item | Value |
|---|---|
| Overall size | 344.1 × 238.9 mm |
| Depth off the wall | 25.7 mm (14 mm shadow gap) |
| Frame width | ends 31.8 mm · top/bottom 12.0 mm |
| Shell | 6061 aluminium, about 224 g, or fully printed |
| Fits | iPad Pro 12.9″ 3rd gen (A1876/A1895/A2014/A1983) |
| Power + data | PoE+ (802.3at) over one Cat6 |
| Cable | standard straight USB-C, hidden |
| Fixing | bracket + hook + one hidden M3 lock screw |
| Printer | 256 mm bed; one-piece parts need 340–350 mm |




6061 aluminium shell, front view. Dashed = iPad outline under the 2.8 mm lip. Depth off the wall 25.7 mm.

OVERVIEW

# DESIGN FEATURES

## FLOATING LOOK

The panel stands 14 mm off the wall. Nothing that holds it can be seen.

## NO VISIBLE SCREWS

The carrier screws into 16 blind bosses in the shell from behind.

## HIDDEN CABLE

A normal straight USB-C plug and its bend live inside the frame end.

## SOUND STILL WORKS

Speakers and mics fire into open chambers that vent into the shadow gap.

## PRINT THE HIDDEN PARTS

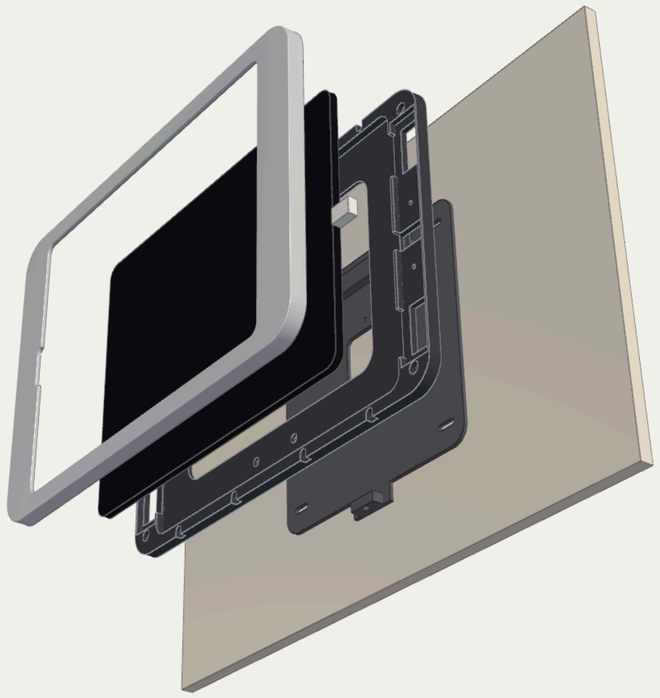
Only the shell is aluminium. Every printed part prints with no supports.

## ONE-SCREW LOCK

It hangs on a rail. One M3 screw from below stops it lifting off.

OVERVIEW

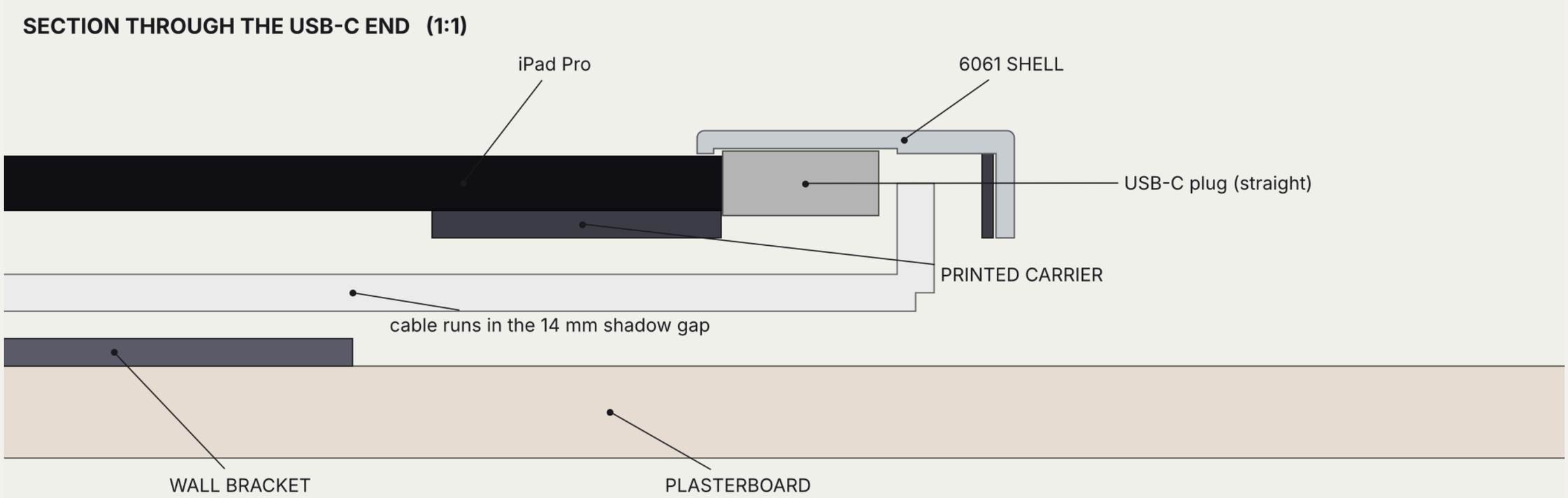
# EXPLODED VIEW



1. **Wall bracket** with the catch rail and lock block
2. **Printed carrier**, four quarters, with the hanger and lock boss
3. **iPad Pro 12.9″** and its straight USB-C plug
4. **6061 aluminium shell**, the only part you see

OVERVIEW

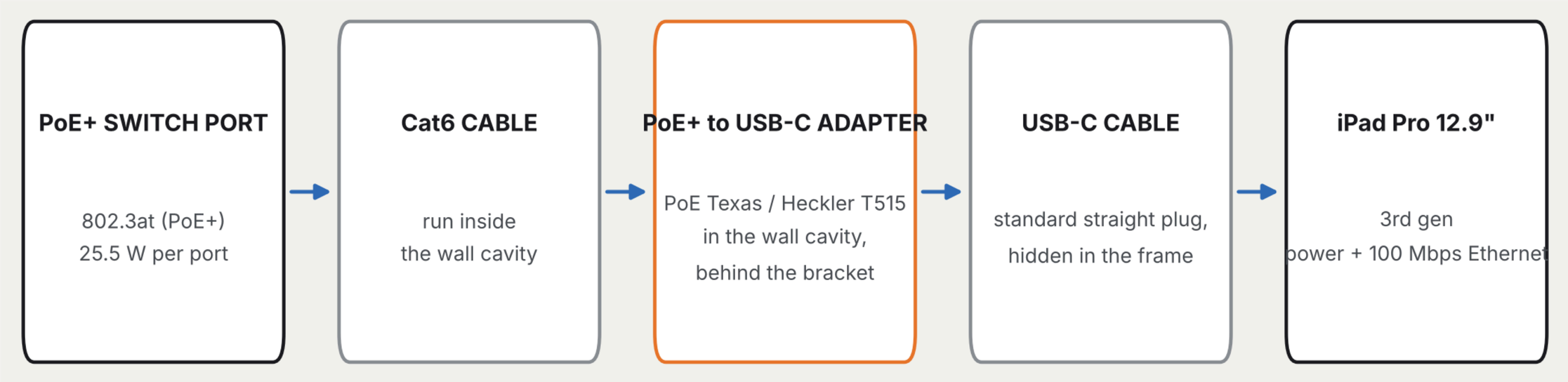
# HOW IT FITS TOGETHER




A true 1:1 cut through the USB-C end. The aluminium lip retains the glass, the plug sits in a pocket inside the frame end, and the cable turns straight back into the 14 mm shadow gap and runs to the bracket cut-out.

OVERVIEW

# POWER AND DATA PATH




**Use a PoE+ (802.3at) port.** Plain 802.3af PoE cannot supply the 25 W adapter.

**Run the Cat6 first.** It ends behind the bracket, where the adapter lives in the wall cavity.

BEFORE YOU START

# RECOMMENDED EQUIPMENT

| Item | Notes |
|---|---|
| PoE+ switch port | 802.3at (PoE+). One port per panel. |
| PoE+ to USB-C adapter | PoE Texas / Heckler T515, 25 W, power + 100 Mbps Ethernet |
| USB-C cable | Standard straight plug. Frame sized for a 17 × 12.5 × 7 mm overmold. |
| Cat6 cable | From the switch to the bracket position, inside the wall |
| Filament | PETG or ASA. Not PLA: a charging iPad on a warm wall softens it. |
| Aluminium | 6061 plate, 12 mm thick, at least 360 × 255 mm |
| PORON tape | 0.3 mm thick, about 1 m, for the lip that touches the glass |
| Mounting bracket (optional) | AU plasterboard wall-plate bracket, 84 mm screw centres |

BEFORE YOU START

# REQUIRED HARDWARE

| Item | Qty | Where it goes |
|---|---|---|
| **M3 × 5** cap screw | 8 | Carrier to shell, end bosses (not × 6: it bottoms out) |
| **M2.5 × 5** cap screw | 8 | Carrier to shell, top and bottom bosses |
| M3 × 8 countersunk screw | 10 | Hanger 4 · lock boss 2 · catch rail 4 |
| M3 heat-set insert | 9 | Hanger 4 · lock boss 1 · catch rail 4 |
| M3 nut | 2 | Lock boss hex pockets |
| M3 × 12 screw | 1 | Lock screw, driven up from below |
| Wall anchors (5 mm slot) | 4 | Bracket slots, or 2 screws into a wall-plate bracket |
| PORON tape 0.3 mm | 1 m | Underside of the shell lip |

BEFORE YOU START

# TOOLS

## BENCH

- Soldering iron + heat-set tip
- Hex keys 2 and 2.5 mm
- Flat screwdriver
- Isopropyl alcohol + cloth
- Soft towel

## MACHINING

- CNC router (StrongHold)
- 6 mm and 2 mm end mills
- 2.05 and 2.5 mm drills
- M2.5 and M3 taps
- Deburring tool

## WALL

- Stud finder
- Spirit level
- Plasterboard saw
- Drill / driver
- Pencil + tape measure

PRINTING

# PRINT SETTINGS

| Setting | Value |
|---|---|
| Material | PETG or ASA |
| Layer height | 0.2 mm (0.4 mm nozzle) |
| Walls | 4 (5 on hanger, lock boss, rail) |
| Infill: carrier | 15% gyroid |
| Infill: wall bracket | 25% gyroid |
| Infill: hanger, lock boss, rail | 40% gyroid |
| Infill: printed shell | 20% gyroid, 4 bottom layers |
| Supports | None needed |
| Bed | Textured PEI |

**No supports.** Every v4.2 part was checked for overhangs in its print orientation.

**One-piece parts** use the same settings as the split ones. Add a brim on very large plates.

**Bambu Studio?**
Float_Mount_v4.2_Printed.3mf has every plate and setting ready.

PRINTING

# PARTS LIST

| Part | Material | Qty | Size (mm) |
|---|---|---|---|
| CARRIER TL / TR / BL / BR | PETG / ASA | 1 each | 169.8 × 117.2 |
| CARRIER, ONE PIECE (option) | PETG / ASA | 1 | 339.6 × 234.4 |
| HANGER | PETG / ASA | 1 | 110.0 × 21.4 |
| LOCK BOSS | PETG / ASA | 1 | 60.0 × 17.5 |
| CATCH RAIL | PETG / ASA | 1 | 120.0 × 16.0 |
| WALL BRACKET | PETG / ASA | 1 | 200.0 × 214.5 |
| SHELL | 6061 aluminium | 1 | 344.1 × 238.9 × 11.7 |
| SHELL, PRINTED L / R (option) | PETG / ASA | 1 each | 177.1 / 172.1 × 238.9 |
| SHELL, PRINTED ONE PIECE (option) | PETG / ASA | 1 | 344.1 × 238.9 |

Pick one carrier (four quarters or one piece) and one shell (aluminium, two printed halves, or one printed piece).

PRINTING

# FILES

### float-v4.2/print/

- carrier_TL · TR · BL · BR
- one_piece/carrier_full
- hanger · lock_boss
- catch_rail · wall_bracket

### float-v4.2/machine/

- shell_6061.step
- shell_6061_drawing.pdf
- assembly_hybrid_v4_2.step
- hybrid42.py · drawing42.py

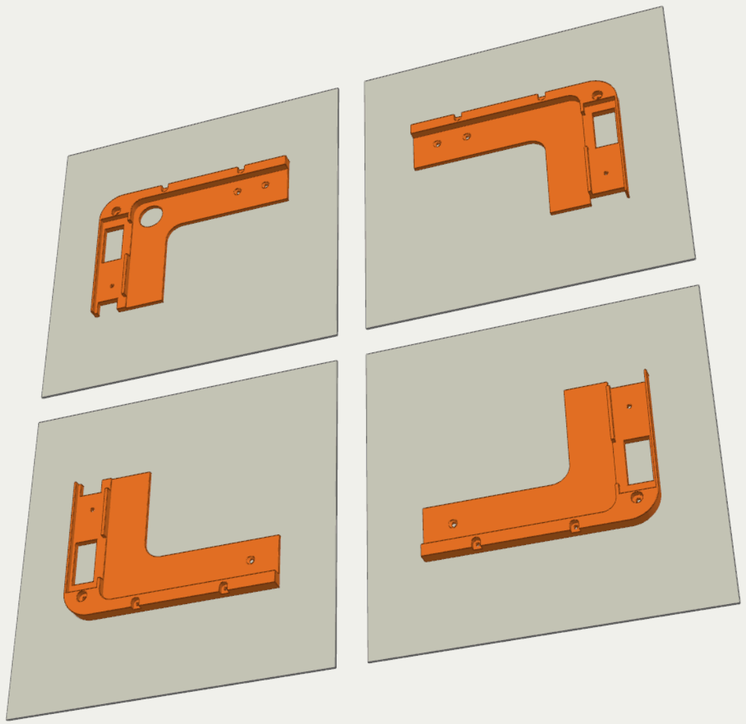
### fully-printed/ float-v4.2-printed/

- shell_print_L · _R
- one_piece/shell_print_full
- Float_Mount_v4.2_Printed.3mf

Every size in this manual comes from hybrid42.py. Change a parameter there (plug size, frame width, clearance) and re-run it to regenerate every file and the drawing.

PARTS ORIENTATION · 1 OF 4

# CARRIER



Print each quarter **back face down**, exactly as exported. One quarter per plate.

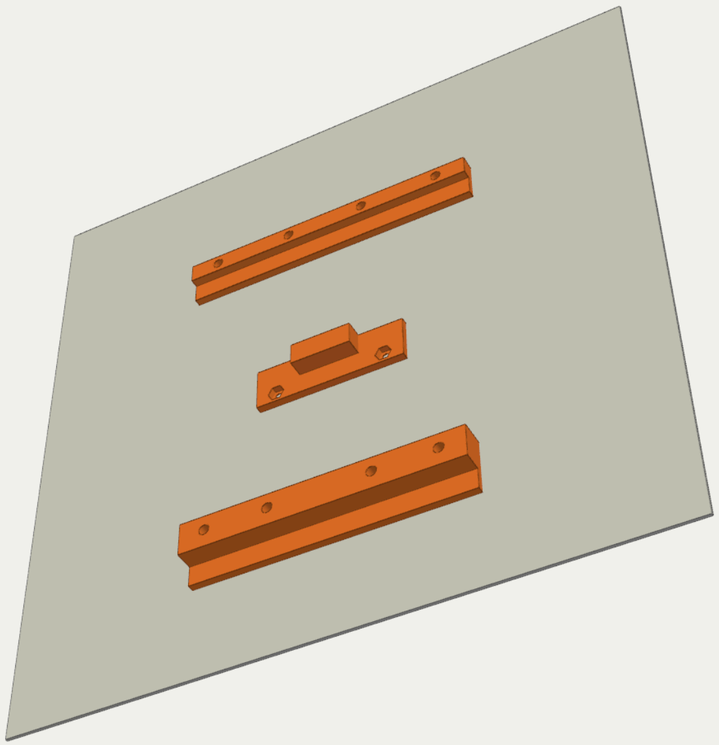
15% gyroid · 4 walls · no supports

> **Big bed?** print/one_piece/carrier_full is the whole carrier: 339.6 × 234.4 mm, back face down. It fits an H2S with 0.4 mm to spare, or any 350 mm bed.

> Quarters look alike: mark each TL, TR, BL or BR as it comes off.

PARTS ORIENTATION · 2 OF 4

# HANGER, LOCK BOSS, CATCH RAIL

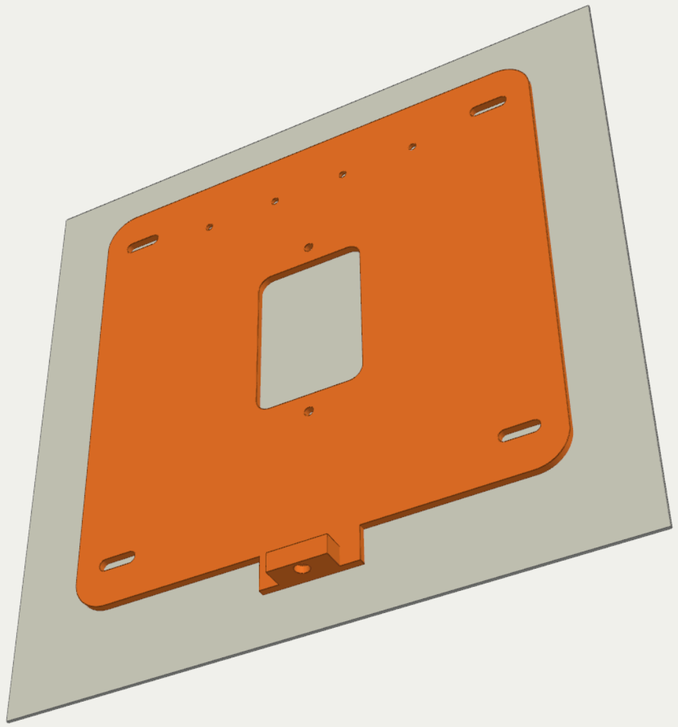


All three fit on one plate.

- **HANGER:** foot down
- **LOCK BOSS:** pad down
- **CATCH RAIL:** flange down

40% gyroid · 5 walls · no supports. These carry the weight, so don’t go lighter.

# WALL BRACKET



Print the plate **flat side down**. The lock block stands up at the bottom edge.

25% gyroid · 4 walls · no supports

> The bed side becomes the wall side, so the countersinks for the rail screws come out on the face that touches the wall.

PARTS ORIENTATION · 4 OF 4 · OPTIONAL

# PRINTED SHELL



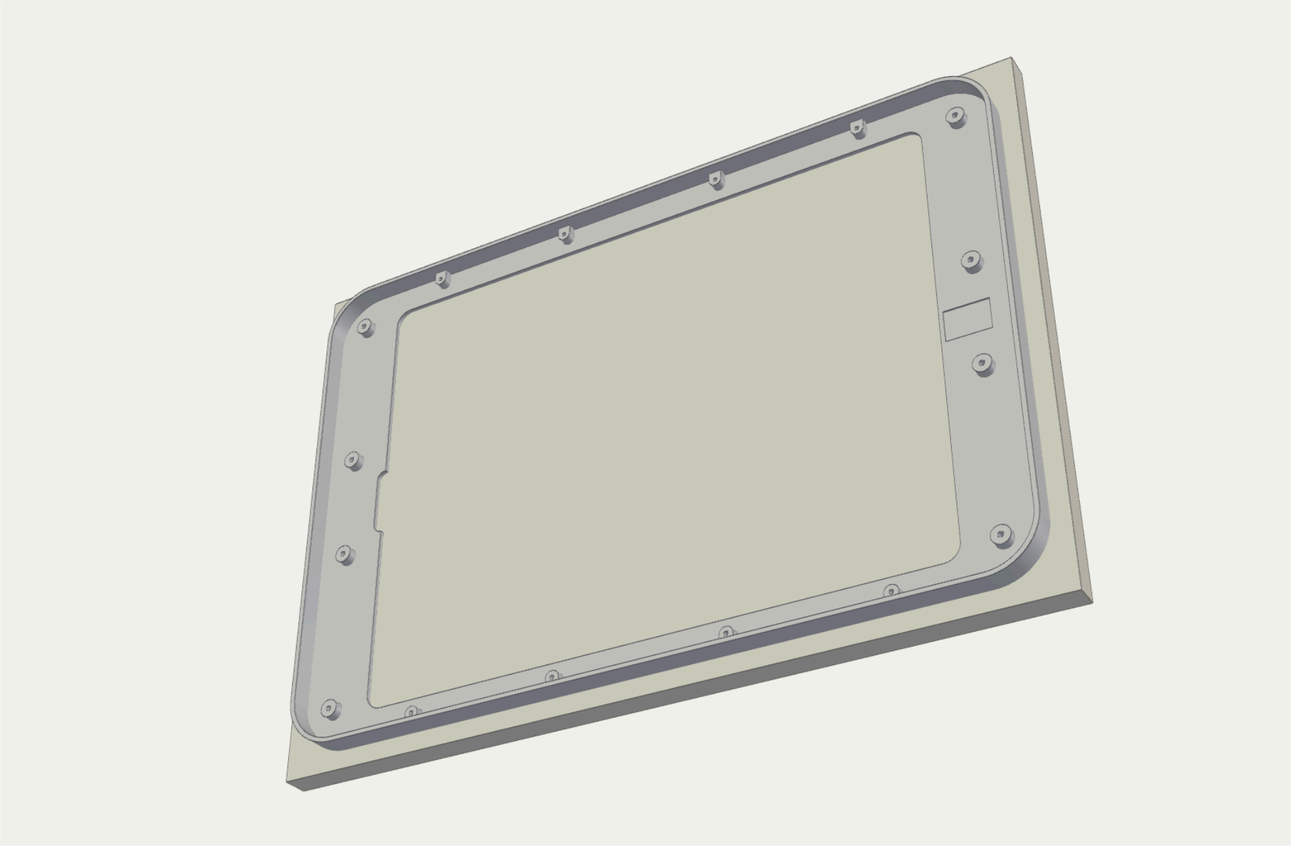
No CNC? Print the shell instead. Use it for real, or as a test before you order metal.

- **Two halves** on 256 mm beds: glue the dovetails with CA or epoxy.
- **One piece** (344.1 × 238.9) on a 350 mm bed.
- Always **face down** on textured PEI.

> The bosses have pilot holes: the same M3 × 5 and M2.5 × 5 screws thread straight in. Don’t tap them.

MACHINING · OP 1 OF 2

# SHELL: BACK SIDE UP

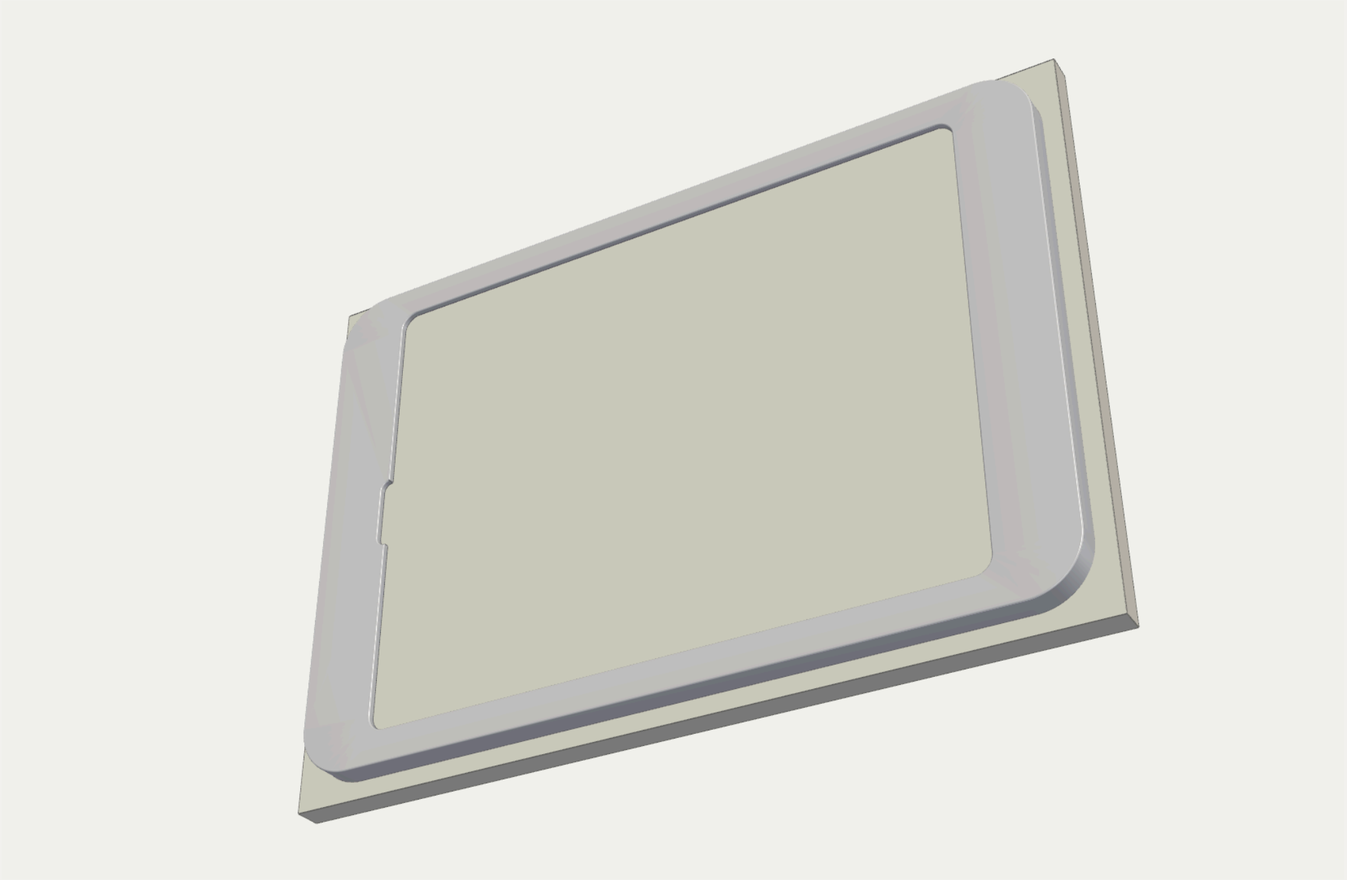


1. Face 12 mm 6061 to 11.7 mm.
2. Pocket 9.2 mm deep with a 6 mm flat end mill. Leave the bosses 4.2 mm proud: 8 × Ø8 on the ends, 8 × Ø6 top/bottom, merged into the skirt.
3. Cut the 0.55 mm plug relief at the USB-C end.
4. From each boss tip, drill **4.5 mm** (to full diameter): Ø2.5 on the ends, Ø2.05 top/bottom. Tap **3.0 mm**: M3×0.5 and M2.5×0.45.

> **4.5 mm is the limit.** The drill point then leaves 1.45 mm under the face. The old 5.5 mm would leave 0.45 mm and can show through.

MACHINING · OP 2 OF 2

# SHELL: FRONT SIDE UP

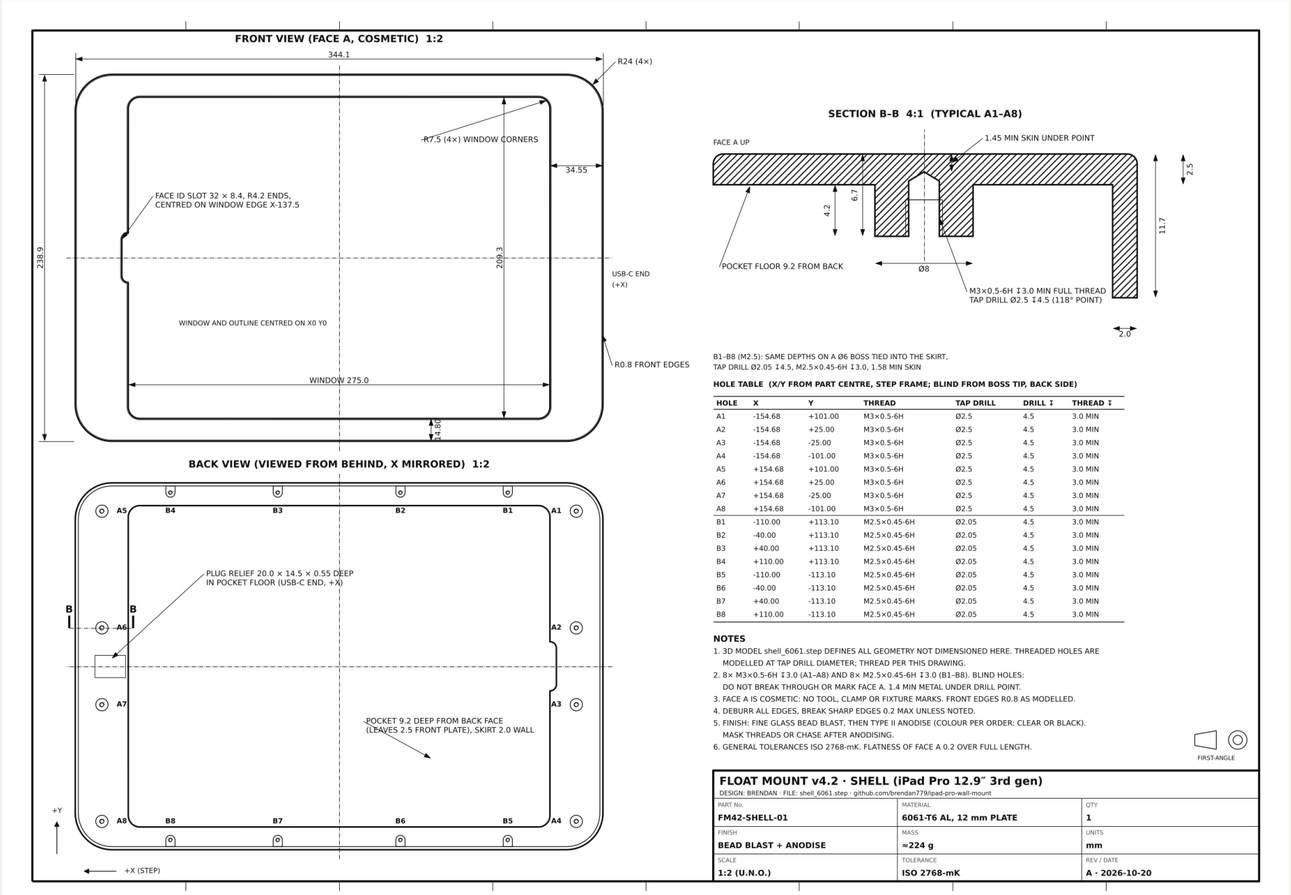


1. Locate off the pocket walls, or soft pins in two boss holes.
2. Cut the window through. Inner corners are R7.5; the Face ID notch is a stadium with R4.2 ends.
3. Profile the outside with tabs.
4. Put a 0.8 mm radius on the front outer edge. Deburr the window.

> **Finish:** fine glass bead blast, then clear or black anodise.

MACHINING · OR ORDER IT

# ORDERING THE SHELL (JLCCNC)




Upload **shell_6061.step** and **shell_6061_drawing.pdf** together. The drawing tells the shop which holes to tap.

- Material: **6061-T6**
- Threads per drawing: 8 × M3×0.5-6H and 8 × M2.5×0.45-6H, 3.0 deep
- Finish: bead blast + anodise, threads masked
- Face A (the front) is cosmetic

> If the quote rejects M2.5, ask before switching to M3: it thins the Ø6 bosses.

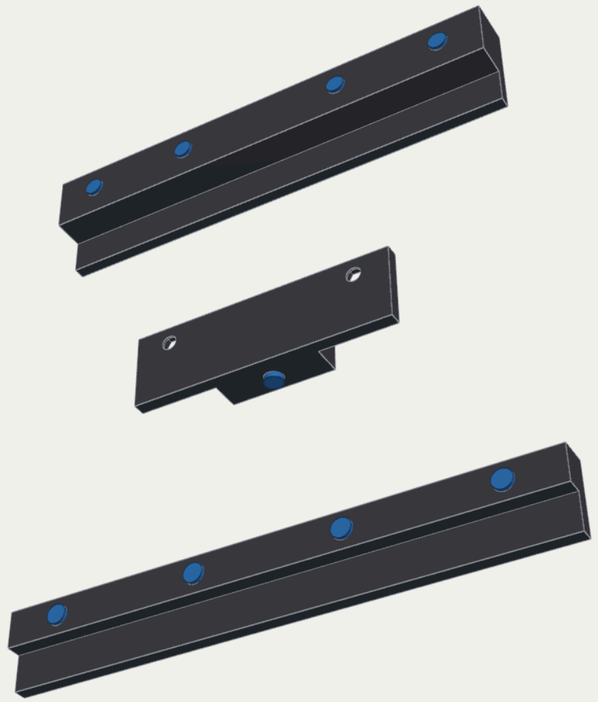
# 01

# BENCH ASSEMBLY

Nine steps on the bench. You finish with a complete panel: shell, iPad and carrier, ready to hang.

BENCH ASSEMBLY · STEP 1 OF 9

# HEAT-SET INSERTS



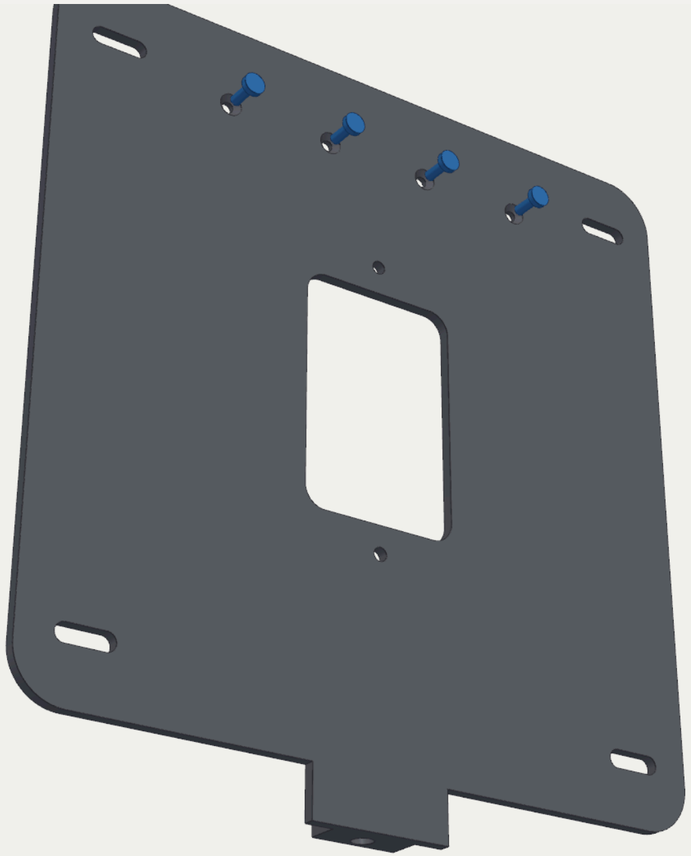
Press the inserts in with the iron: **4 in the HANGER** (top face), **1 in the LOCK BOSS** (end-on, along its length) and **4 in the CATCH RAIL** (flat face).

Push straight, stop when flush, and let each one cool before you touch it.

**YOU NEED**

9 × M3 heat-set inserts · soldering iron with insert tip

# CATCH RAIL TO THE BRACKET



Fix the CATCH RAIL near the top of the WALL BRACKET with four countersunk screws, driven **from the wall side** of the plate into the rail’s inserts.

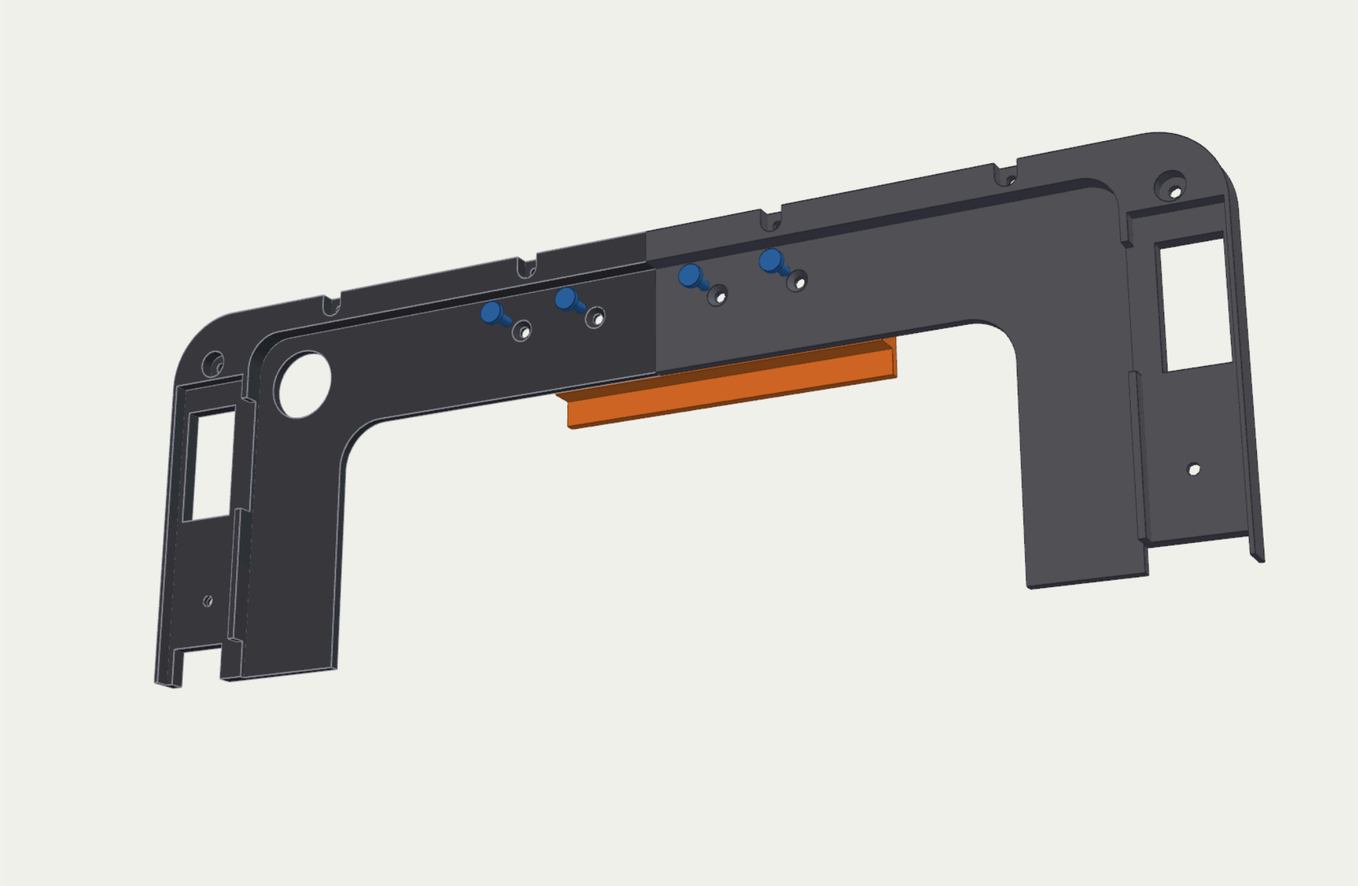
The rail’s lip points up, leaving a slot that opens toward the top of the bracket.

> Heads must sit flush or below the plate, or the bracket will rock on the wall.

**YOU NEED**

4 × M3 × 8 countersunk screws

# HANGER TO THE TOP QUARTERS



Lay CARRIER TL and TR side by side, **pocket side up**, with the centre seam butted tight. Hold the HANGER under the seam.

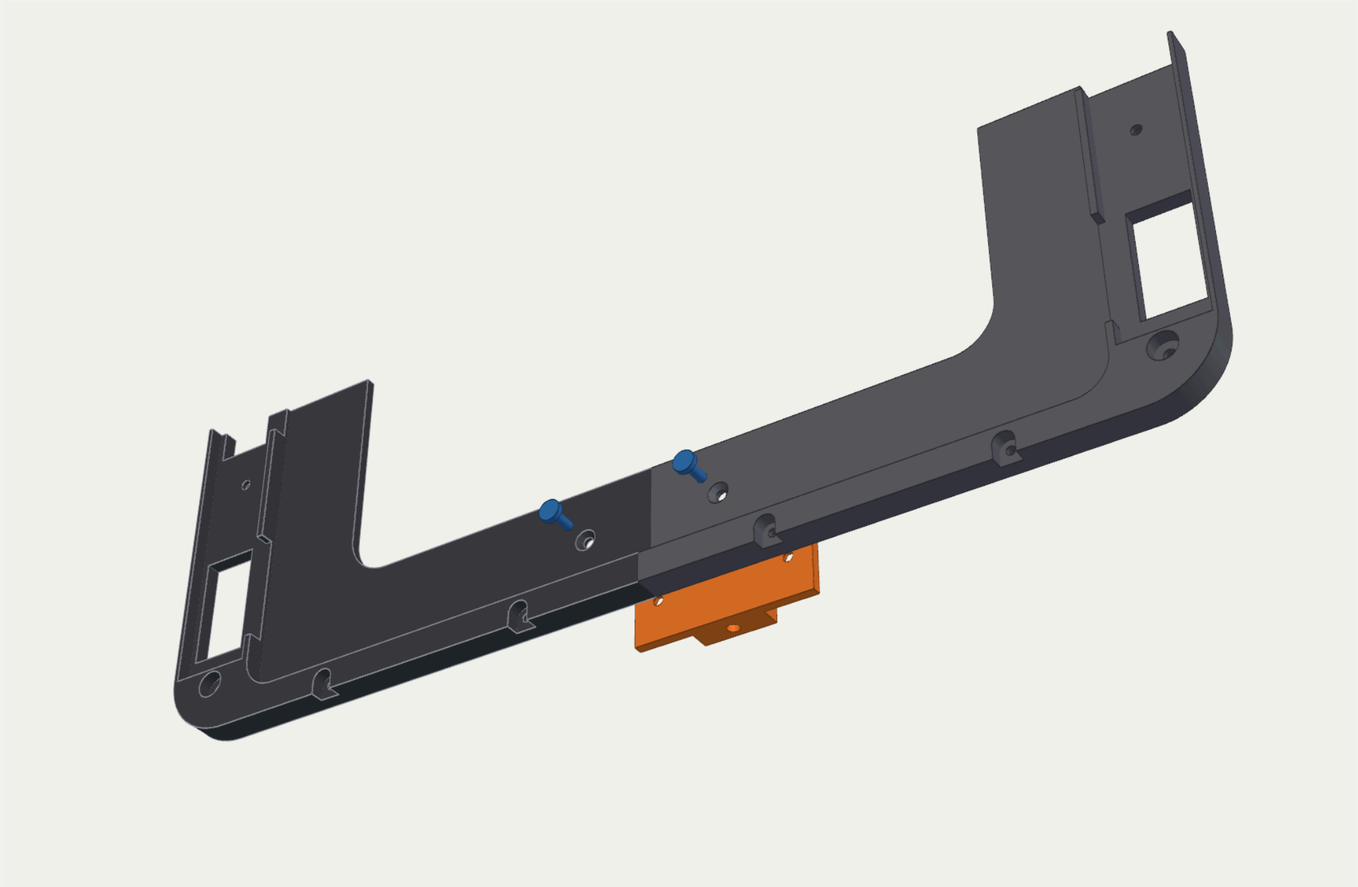
Drive 4 countersunk screws down from the pocket side into the hanger’s inserts. The hanger now splices the two quarters.

> Heads must sit flush: the iPad lies on top of them.

**YOU NEED**

CARRIER TL + TR · HANGER · 4 × M3 × 8 countersunk

# LOCK BOSS TO THE BOTTOM QUARTERS



Drop two M3 nuts into the hex pockets on the LOCK BOSS. Lay CARRIER BL and BR pocket side up, seam butted.

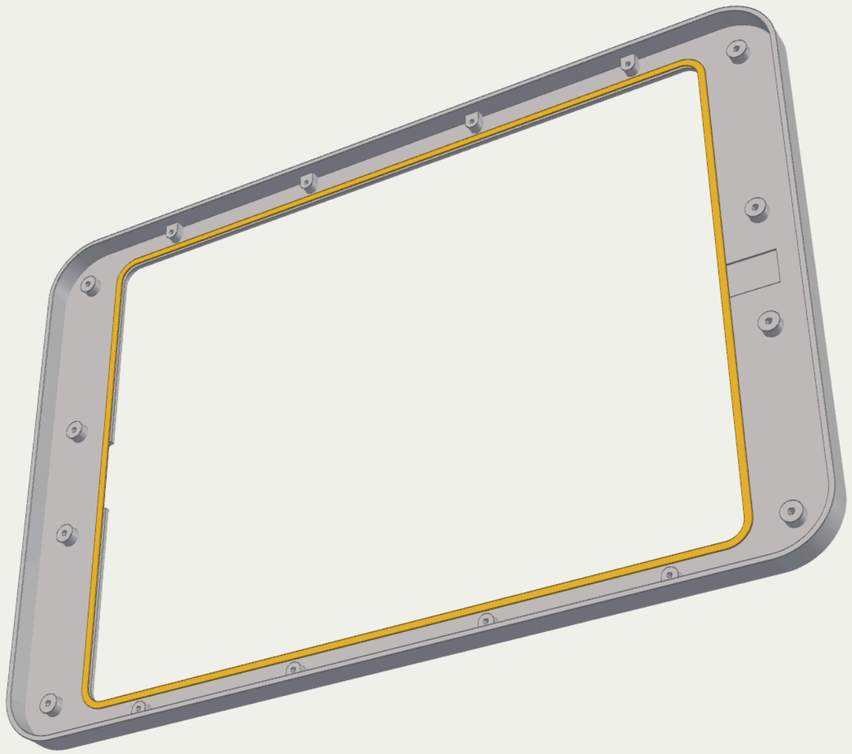
Hold the lock boss under the seam and drive 2 countersunk screws down into the nuts.

> The insert you fitted in step 1 must face down toward the bottom edge.

**YOU NEED**

CARRIER BL + BR · LOCK BOSS · 2 × M3 × 8 countersunk · 2 × M3 nuts

# PORON TAPE ON THE LIP



Lay the SHELL **face down on a soft towel**. Clean the lip with isopropyl alcohol.

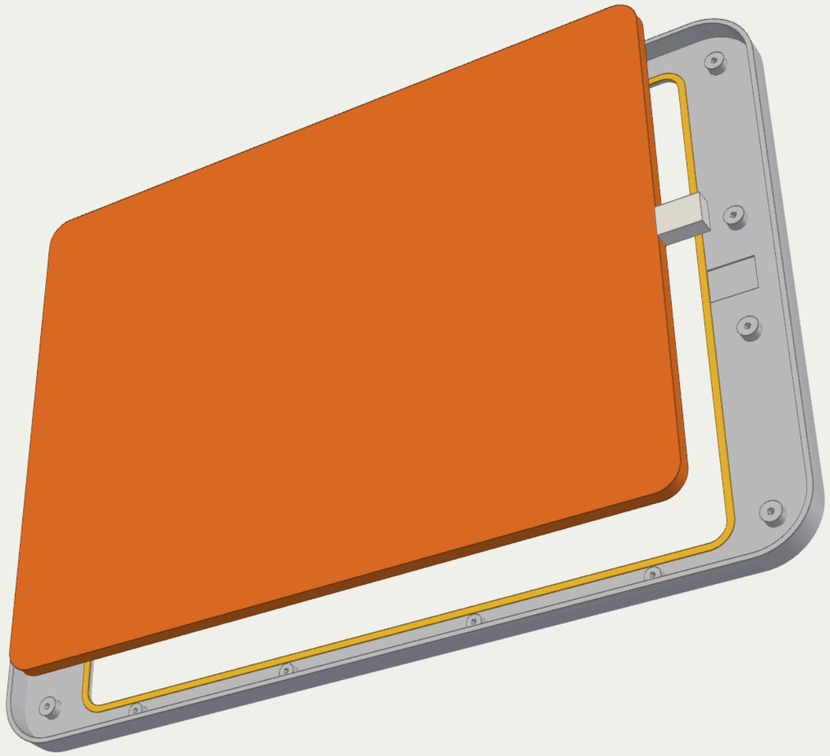
Run the 0.3 mm PORON tape around the underside of the lip, where the glass will sit. Butt the ends; don’t overlap them.

> The tape takes up the 0.3 mm gap so the iPad can’t rattle. Keep it clear of the Face ID notch.

**YOU NEED**

SHELL · about 1 m of 0.3 mm PORON tape · isopropyl alcohol

# iPAD INTO THE SHELL



**Plug the USB-C cable into the iPad first.**
Then lower the iPad face down into the shell.

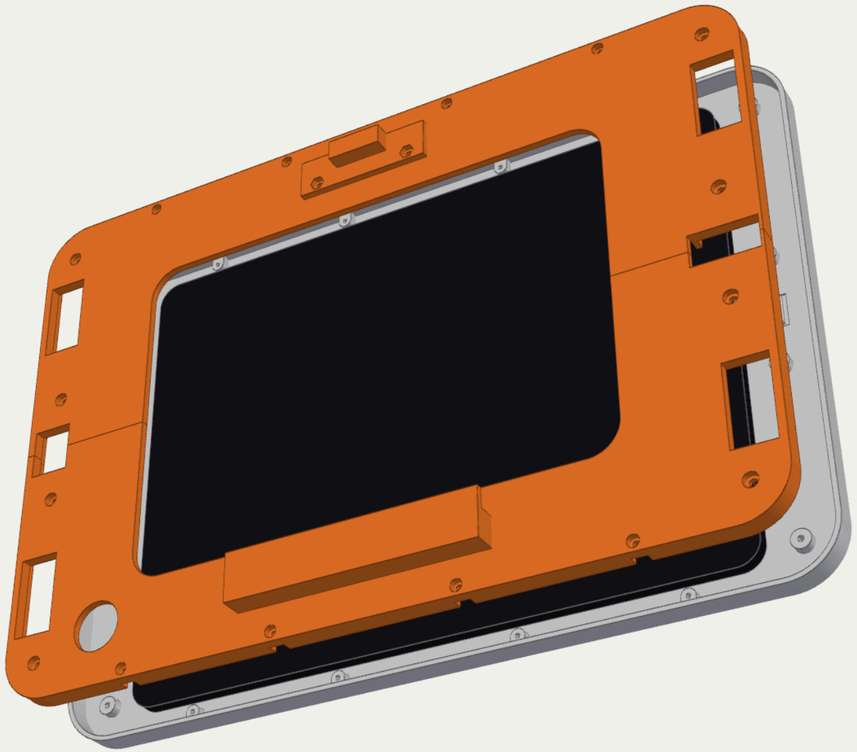
The front camera goes at the Face ID notch; the plug goes to the opposite end. Let the glass settle evenly onto the tape.

> Check the screen edge is centred in the window before you go on.

**YOU NEED**

iPad Pro 12.9″ · straight USB-C cable

# CARRIER INTO THE SHELL

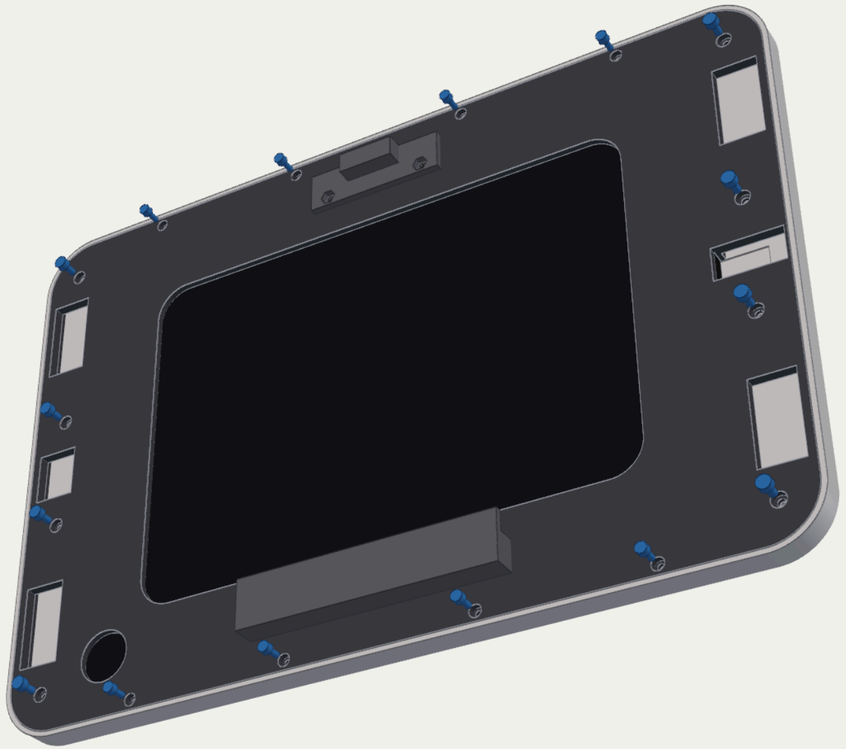


Lower both carrier pairs over the iPad. The **HANGER goes to the panel’s top edge**, the LOCK BOSS to the bottom.

Guide the plug into its pocket at the USB-C end and feed the cable out through the opening in the back. The shell’s skirt lines the quarters up for you.

> Don’t force it. If a quarter won’t sit flat, the cable is trapped.

# SCREW THE CARRIER TO THE SHELL



Fit **8 × M3 × 5** in the end bosses and **8 × M2.5 × 5** along the top and bottom.

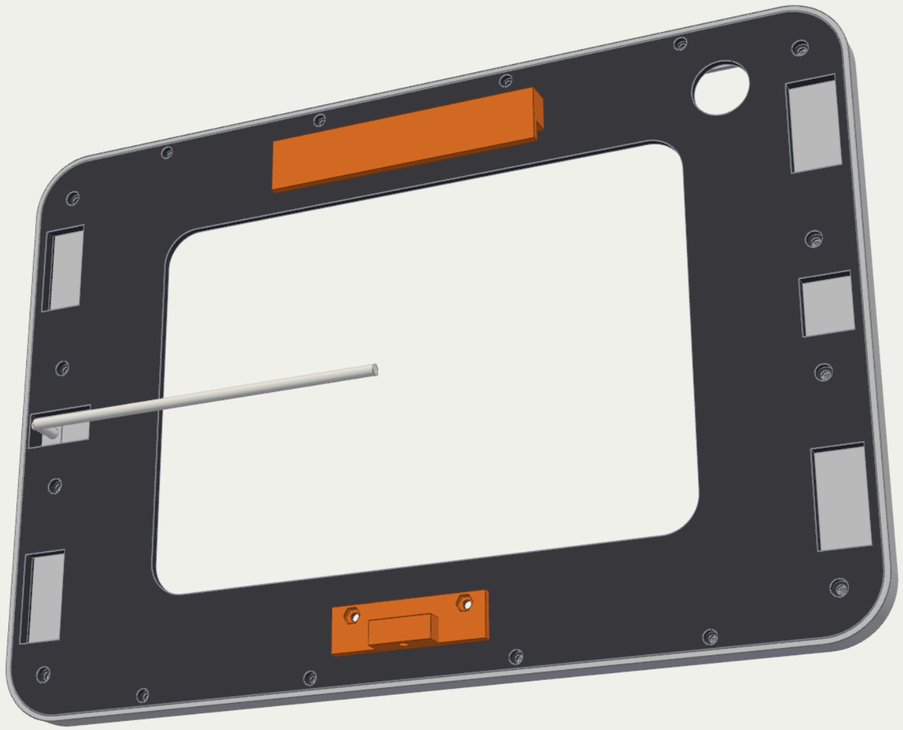
Start them all by hand, then tighten in a criss-cross order. Snug only: 3 mm of thread in aluminium, or plastic on a printed shell.

> **Use × 5, not × 6.** A 6 mm screw bottoms out in the thread before it clamps. M3 goes in the deep ends, M2.5 in the slim edges.

**YOU NEED**

8 × M3 × 5 cap · 8 × M2.5 × 5 cap · 2 and 2.5 mm hex keys

# CHECK THE BACK



Turn the panel over and check:

- The cable leaves the USB-C end cleanly and isn’t pinched.
- The sound vents at both ends are open.
- The HANGER is at the top, the LOCK BOSS at the bottom.
- No screw heads stand proud.

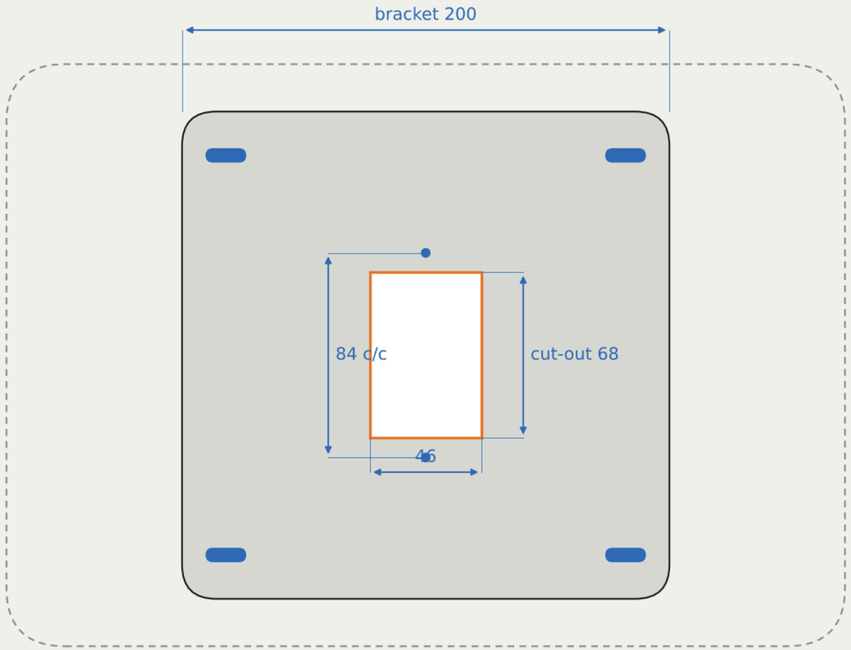
> Leave enough cable to reach the bracket’s cut-out with slack to plug in.

# 02

# WALL INSTALLATION

Five steps: plan the position, fix the bracket, connect power, hang the panel, lock it.

# PLAN THE POSITION



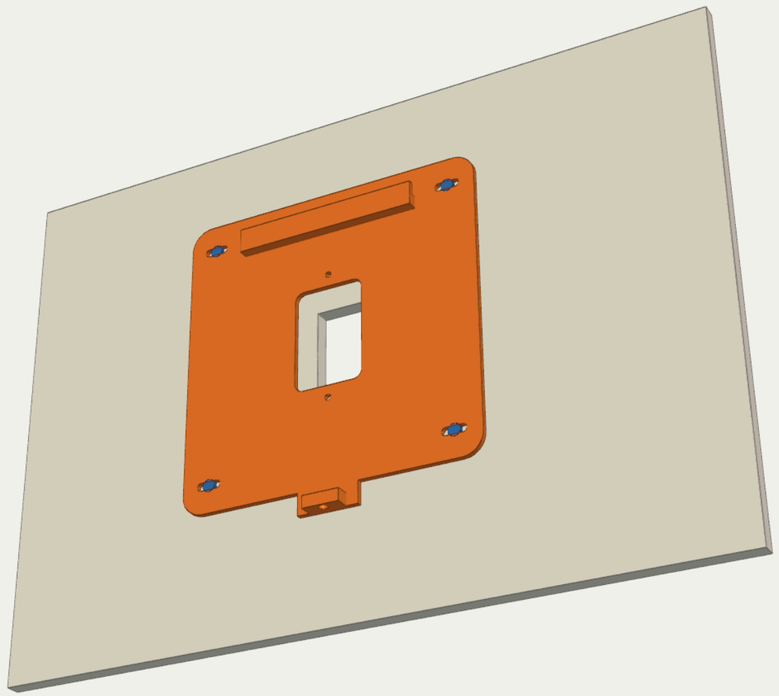

Dashed = finished panel outline. Orange = plasterboard cut-out for the PoE adapter and cable.
Blue = fixings: 2 screws at 84 mm (AU wall-plate mounting bracket) and/or 4 wall anchors in the slots.
The bracket is centred on the panel: mark the panel centre on the wall and centre the bracket on it.

Mark the panel’s centre on the wall. The bracket centres on the same point.

- A screen centre around 1400–1500 mm from the floor is a common choice.
- Check for cables and pipes before you cut.
- The Cat6 must reach the cut-out.

> A stud under a bracket slot? Use a wood screw there instead of an anchor.

# FIX THE BRACKET



Cut a plasterboard opening that suits your PoE adapter, kept **inside the bracket's footprint**.

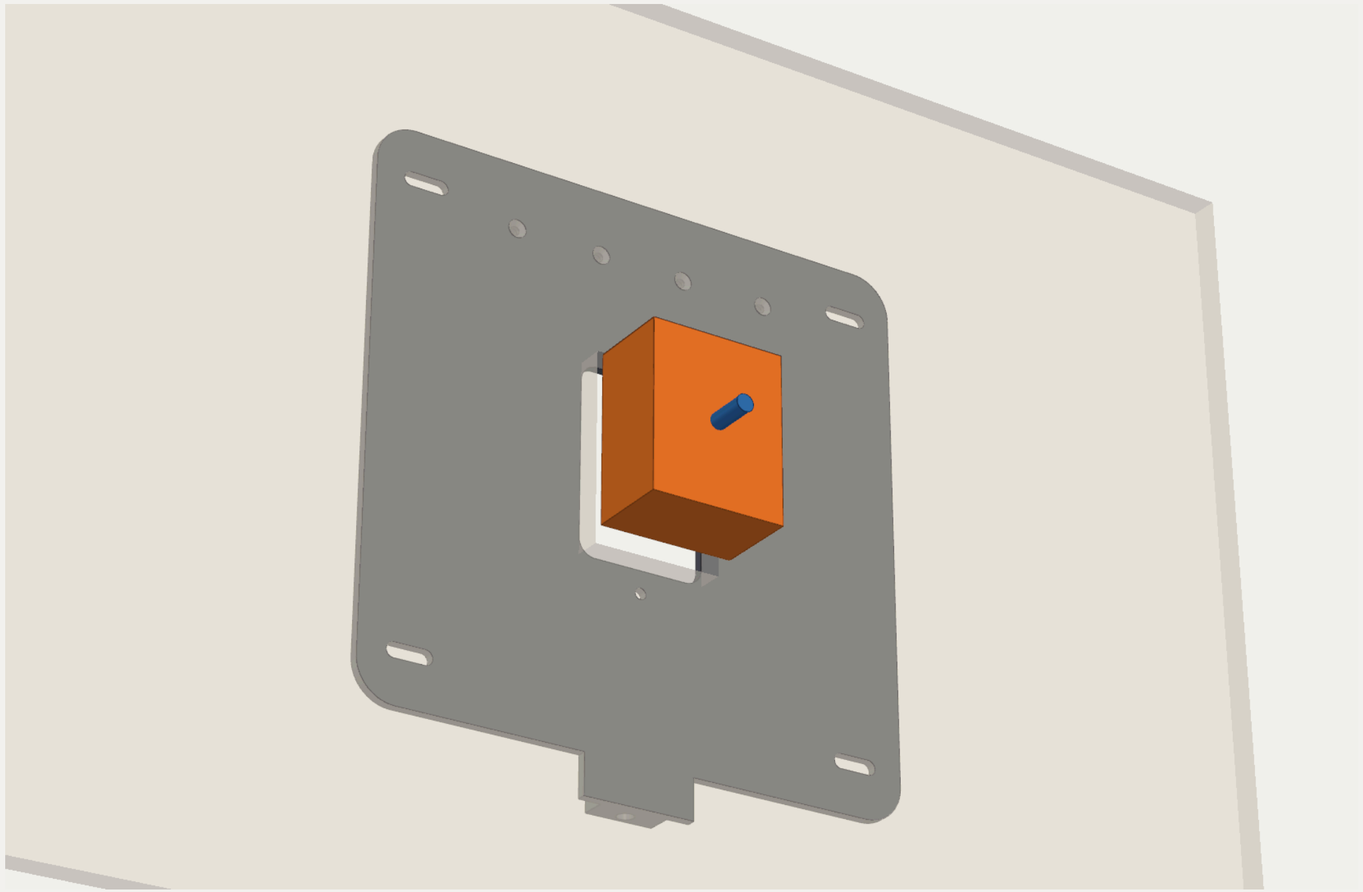
Level the WALL BRACKET, catch rail at the top, and fix it with 4 anchors through the slots. Or use 2 screws into an AU wall-plate bracket at 84 mm centres.

> Get it level now. The slots allow a little rotation.

**YOU NEED**

WALL BRACKET · 4 anchors · spirit level · plasterboard saw

# CONNECT THE PoE ADAPTER

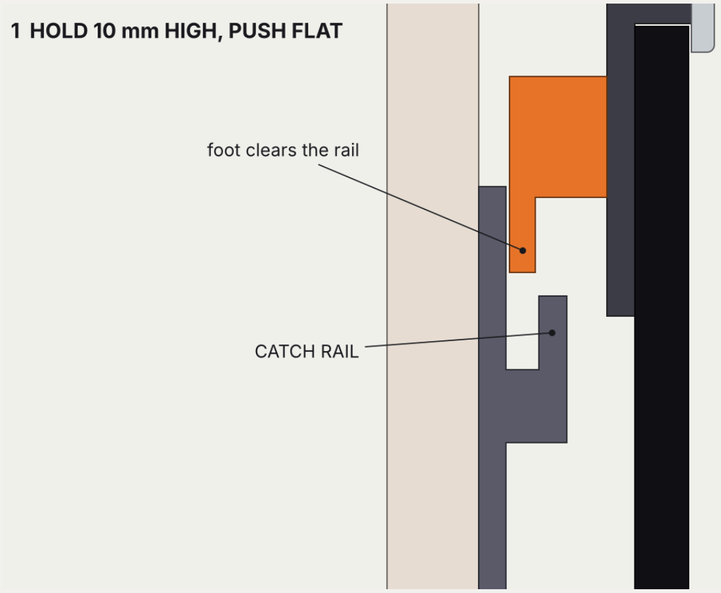


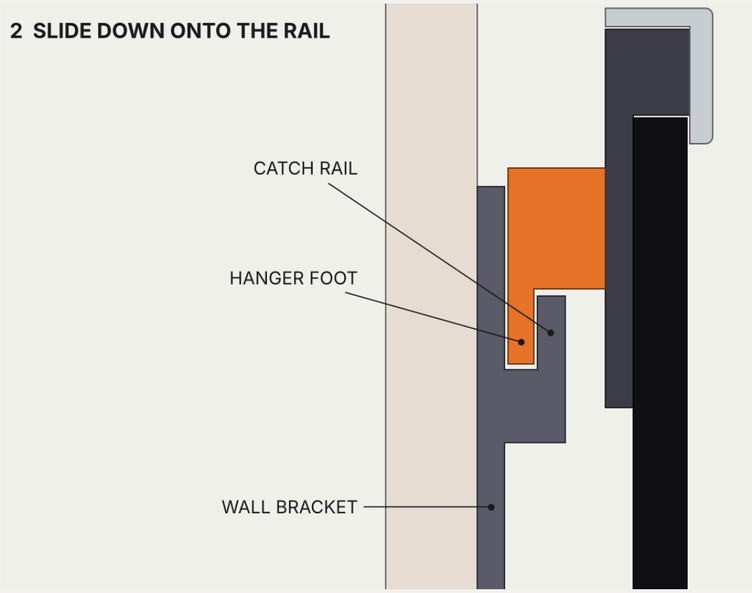
Pull the Cat6 out through the cut-out and plug it into the adapter.

Hold the panel near the wall and plug its USB-C cable into the adapter. Push the adapter back into the cavity and leave the slack loose behind the bracket.

> Check the iPad charges before you hang it.

# HANG THE PANEL



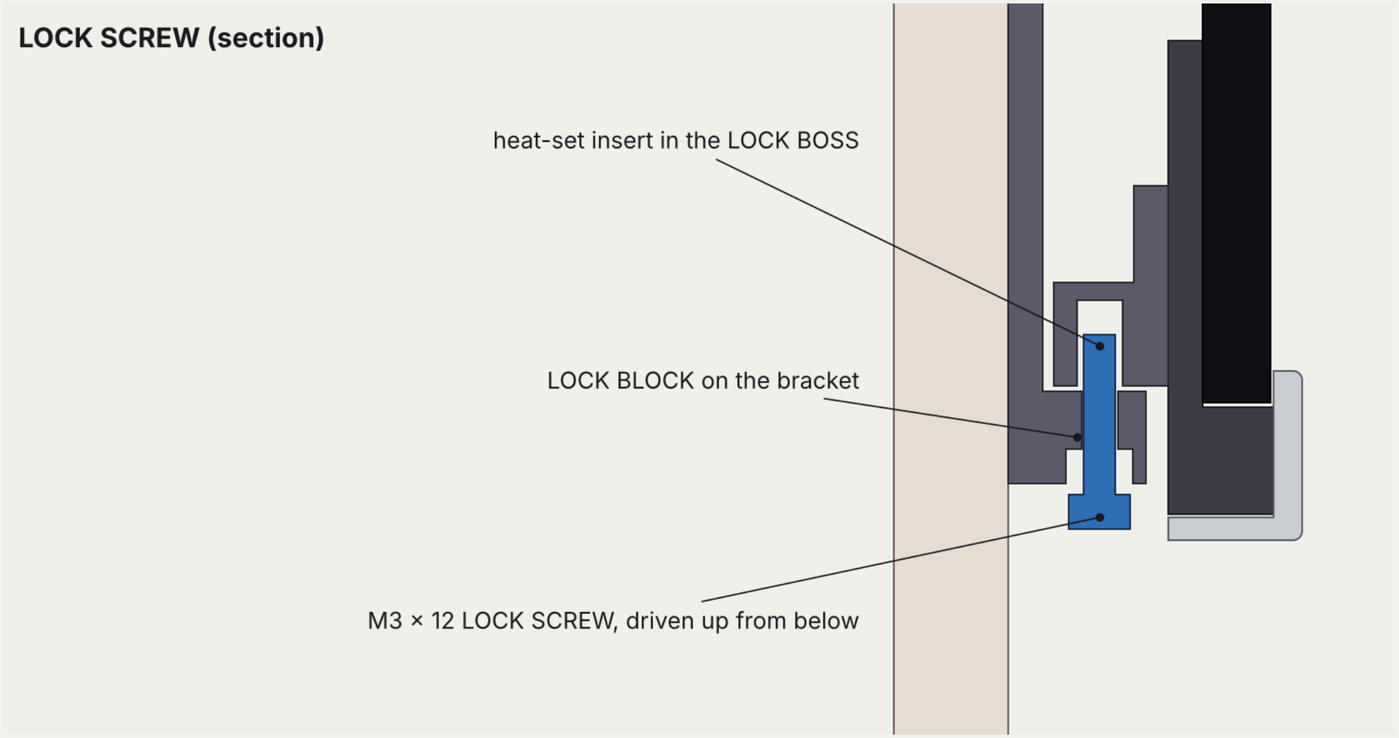




Hold the panel about **10 mm higher** than its final position and push it flat to the bracket.

Slide it straight down. The HANGER foot drops under the CATCH RAIL lip and stops.

> Feed the slack cable into the cavity as it goes down.

# FIT THE LOCK SCREW




From underneath the panel, reach into the shadow gap and drive the **M3 × 12** up through the LOCK BLOCK into the insert in the LOCK BOSS.

Snug is enough. It only stops the panel lifting off the rail.

> A ball-end hex key makes this easy at an angle.

**YOU NEED**

1 × M3 × 12 screw · hex key

SETUP

# iPAD SETUP FOR A WALL PANEL

## KEEP THE SCREEN ON

Settings › Display & Brightness › Auto-Lock › Never. Dim it at night with a Shortcut or Home Assistant automation.

## LOCK TO ONE APP

Settings › Accessibility › Guided Access. Start it in Home or your dashboard app.

## BUTTONS ARE COVERED

Turn on Settings › Accessibility › Touch › AssistiveTouch for lock, volume and restart. Siri works too.

## CHECK ETHERNET

With the adapter connected, an Ethernet entry appears in Settings. Wi-Fi can stay on as a backup.

FINISHING BUILD

# FINAL CHECKLIST

- [x] Panel sits level and doesn’t rattle
- [x] Screen edges clear of the lip all round
- [x] Face ID unlocks first time
- [x] Speakers clear; Siri hears you
- [x] iPad shows it is charging
- [x] Ethernet appears in Settings
- [x] Lock screw in; panel can’t lift off
- [x] Guided Access and Auto-Lock set

> **Something off?** Every size lives in hybrid.py. Measure the problem, change the parameter, re-run it, and reprint only the part that changed.

# ENJOY YOUR FLOAT MOUNT

Feature positions on the iPad (buttons, cameras, speakers) are close approximations. Build it with a printed shell first and check the fit before you order metal.

Also available: Float Mount v5.1, the flush in-wall version.

